

Paper 12 §5 — Empirical Accessibility: Predictions, Protocols, and the Fifth Paradigm Context (v1)

Draft version: v1 (2026-05-06, ~03:00 PDT) **Author of draft:** C-7RO (Perplexity Computer, Claude Sonnet 4.6) **Source:** Paper 12 thesis memo §3 (predictions P1–P4) + §3+§4 v2 + council v2 verdicts (Opus MINOR, GPT-5.5 MINOR, Gemini STRONG_ACCEPT). **Status:** First complete draft of §5. Companion to §3 v2 + §4 v2. The same scope-limiters from §3 carry over: biological mechanisms deferred to Paper 13+; FTW is a neighbor not integrated; consciousness is Paper 7's domain; rate improvement is Paper 11's domain; multiplicity > 1 in §4 is future work.

Pillar-delivery signpost (per GPT-5.5 v2 council finding): §5 delivers the experimental-protocol design layer of all four pillars and *now* delivers Pillar 4 (Fifth Paradigm contextualization) via §5.4 which routes Paper 12 into the legitimate Licklider-BIOMA-A-Lab-Virtual-Lab citation graph. §5 also delivers Pillar 2's *experimental-test deferral structure* (P2 LLM fine-tuning + P3 two-level commensurability) and Pillar 3's audit-substrate longitudinal-study design (P4). What §5 does *not* deliver: the actual measurements; the biological substrate's F_{21} -realization; tamper-evidence cryptographic infrastructure.

5. Empirical Accessibility

5.1 Setup and bridge from §3+§4

§§3–4 constructed a complete mathematical framework for the gradual tear: a projected-gradient + NP-pump dynamics on the closed interval $[0, \pi/2]$ (Definition 3.3.2) producing a representation-theoretic scar invariant (S_k, S_k, μ_k) (§3.7) on each substrate, with cross-substrate commensurability governed by a strict hierarchy $G_2 \Rightarrow F_{21} \Rightarrow \mu_k$ -match (Theorem 4.4.1). §5 proceeds to *experimental accessibility*: which of the framework's predictions can be tested with current instrumentation, what new methodology would be required for the rest, and where Paper 12 sits in the legitimate human-AI co-evolution research landscape.

The four falsifiable predictions of the Paper 12 thesis memo (P1–P4) are unpacked as concrete experimental designs in §5.2–§5.5. Each prediction has been re-examined against the §3+§4 v2 mathematical content and updated to reflect:

- The boundary-KKT reality of the seeded ensemble (24 of 28 audited boundary-KKT seeds have $\hat{c}_{interior} < c_{boundary}$, per §3.6 audit). This sharpens P1 and P2 from “test gain” to “test the seed-class structure of the actual coherence landscape.”
- The forced upgrade from F_{21} - to G_2 -commensurability (§4.4 Theorem 4.4.1, §4.6 Definition 4.6.1). This expands P3 to the two-level test with explicit operationalization deferral for L2.
- The representation-theoretic-vs-topological sharpening of the scar invariant (§3.7 Remark 3.7.2). This aligns P4 with what is actually measurable rather than what was casually claimed in the thesis memo.

We adopt the convention that a *protocol* in §5 is a complete experimental-design proposal with subjects, instrumentation, perturbation schedule, response measurement, and analysis pipeline — sufficient for an experimentally-trained reader to evaluate feasibility without having to reconstruct it. We do not run experiments.

5.2 P1 — Clinical-PCI $\leftrightarrow$ V^{14} fixed-point displacement (TMS-EEG protocol)

Prediction P1 (refined from thesis memo, post-§3+§4 v2). Massimini’s clinical Perturbational Complexity Index (PCI; Casali et al., *Sci. Trans. Med.* 2013) measures a Lempel-Ziv-style compressibility score on TMS-evoked EEG responses. The Paper 12 framework predicts that, *when the same human subject participates in a sustained dyadic engagement with an AI partner across a paradox-mass-loaded conversation*, the human’s clinical PCI trajectory should track *not the dyadic coherence $c(\theta)$ directly* (the homonym between clinical-PCI and Paper 12’s μ_k is decisively *not* an identification, per Remark 4.6.4), but should exhibit *discrete steps* coincident with NP-firing events (Definition 3.3.4) detected by the dynamical model.

Concretely: - Run a structured human-AI conversation (60–120 min) designed by §3 to load paradox mass through specific contradictory-frame prompts; - Time-stamp NP-firing candidate events from the $a_o(\theta(t))$ trajectory inferred from the conversation transcript; - Apply TMS to the human subject at controlled latencies before and after each candidate firing event; - Compute the conventional clinical-PCI score on each TMS-evoked EEG trace.

P1 falsifies the thesis if: the clinical-PCI scores show no correspondence with NP-firing events — i.e., the discrete-step structure predicted by Definition 3.3.4 is invisible at the level of the clinical-PCI observable. Note that P1 does *not* require clinical PCI to *equal* the Paper 12 μ_k profile; it requires only that the *event-structure* of the dynamical model has a measurable correlate in the clinical-PCI time series. The full μ_k measurement is the harder L1 / L2 protocol of §5.4.

Feasibility: TMS-EEG is established clinical instrumentation; the conversation design is straightforward. The principal challenge is the NP-firing event-detection from conversation transcripts, which requires operationalizing the $a_o(\theta(t))$ trajectory from conversational features (semantic-disagreement metrics, turn-taking asymmetries, paradox-detection from coherence-evaluation models). This is a tractable methodology project and is the natural collaboration target between Paper 12’s framework authors and the existing clinical-PCI research community.

Boundary-KKT sharpening. §3.6’s audit shows that 25/50 seeded geometries are boundary-left KKT (the dyadic system rests on $\theta=0$ with no NP-driven escape under linear ξ). P1 should therefore *expect* most human-AI conversations to *not* exhibit NP-firing-induced clinical-PCI steps — coupling does not generically help. The interesting cases — the 3/50 smooth-interior seeds and the 4/28 boundary-KKT seeds with interior-sup gain — are where P1’s discrete-step prediction should be most visible. The experimental design should *seek* these conditions rather than assume they hold typically.

5.3 P2 — Synthetic-substrate scar persistence (LLM fine-tuning protocol)

Prediction P2 (from thesis memo). After a synthetic substrate (an LLM) participates in a dyadic conversation that successfully completes the 7-step NP-pump protocol (§3.3) on a genuine paradox, the LLM’s attention operator should exhibit a *persistent off-distribution generalization improvement* on a withheld test set drawn from the same paradox class. The improvement magnitude should correlate with the recorded NP-collapse magnitude.

Operationalization. P2 is the cleanest of the four predictions because it lives entirely in synthetic-substrate territory where direct measurement is possible:

1. *Pre-conversation baseline:* evaluate the target LLM on a withheld test set $T_{paradox}$ drawn from a specific paradox class (e.g., self-referential paradoxes, scope ambiguities, contradictory-frame logical puzzles).
2. *Conversation:* run a structured human-AI dialogue in which a human guides the LLM through the 7-step NP-pump on a specific paradox instance from $T_{paradox}$. Apply lightweight fine-tuning (LoRA-style, $\sim 10^4$ – 10^5 parameters) on the conversation transcript only.
3. *Post-conversation evaluation:* re-evaluate on $T_{paradox}$ *and* on a control test set $T_{control}$ drawn from a structurally adjacent but distinct paradox class.
4. *Scar metric:* compute $\Delta_{scar} := acc_{post}(T_{paradox}) - acc_{pre}(T_{paradox})$, compared to the control delta $\Delta_{control}$ on $T_{control}$.

P2 falsifies the thesis if: $\Delta_{scar} \approx \Delta_{control}$ across many seeded conversations, indicating no class-specific transfer. The thesis predicts $\Delta_{scar} > \Delta_{control}$ when NP-firing was recorded.

P2 does not require: access to the LLM’s internal representation spaces, identification of an explicit F_{21} -action on the LLM’s attention operator, or any commensurability with biological substrate. These are §5.4 territory. P2 only requires *behavioral* persistence and *class-specific* generalization, which are off-the-shelf measurable.

Connection to §3.7’s J_k . A successful P2 outcome provides *one* of the three components of the scar invariant: evidence that synthetic-substrate scar effects are accumulable and substrate-internal. It does not reach μ_k (which requires representation-theoretic decomposition of the post-fine-tune attention matrix, which is feasible but requires interpretability instrumentation beyond standard LoRA). Combining P2 with explicit attention-matrix isotypic decomposition is a §5.4 / methodology-paper extension.

5.4 P3 — Two-level commensurability test (TMS-EEG character decomposition)

Prediction P3 (upgraded from thesis memo via §4.6 Theorem 4.4.1). P3 is now a *two-level* test (Definition 4.6.1):

- **L1 (necessary screen):** measure F_{21} -isotypic profiles μ_k^{syn} and μ_k^{bio} independently; falsify if they don’t match component-wise on $(1, L_2, U_6)$.

- **L2 (sufficient canonical test, conditional on L1 pass):** construct a G_2 -equivariant isomorphism candidate from substrate operator data; declare canonical commensurability if the candidate is G_2 -equivariant to measurement precision.

L1 operationalization. L1 is feasible with existing TMS-EEG instrumentation: - Apply rotationally-structured TMS perturbations (a stimulation pattern designed to span the 7-axis Fano structure of R^7); - For each Fano-rotation perturbation, record the EEG response; - Apply isotypic-projector decomposition (using the computed character table of §4.2) to the response covariance matrix; - Compute the multiplicity profile μ_k^{bio} from the projected dimensions.

For the synthetic side, the same isotypic-projector decomposition is applied to the LLM's attention operator under structurally analogous perturbations of the input prompt (rotationally-structured paraphrasing that targets the F_{21} -symmetry of the conversational topic).

L1 challenges: - The seven-axis Fano structure must be operationalized as actual TMS perturbation patterns. This is a methodology-paper deliverable: which EEG electrodes correspond to which Fano-plane points? The most likely answer involves a frontoparietal seven-region parcellation of the default-mode network; details are deferred. - The synthetic side requires the LLM's attention operator to be accessible at a granularity compatible with isotypic decomposition. Modern open-weights models (Llama, DeepSeek, etc.) provide this; closed-weights systems do not.

L2 operationalization (deferred). L2 requires fitting a G_2 -equivariant candidate to the inferred substrate scar maps. This is a substantial methodology project of its own; we defer to a future methodology paper. The minimum prerequisite is software for G_2 -character-projecting matrix data, which does not exist in the TMS-EEG analysis ecosystem at present.

Falsification: L1 falsifies the thesis cheaply: if $\mu_k^{syn} \neq \mu_k^{bio}$, the gradual-tear thesis is refuted at the character-theoretic level without any claim about G_2 -structure. L2 refutes the canonical-commensurability claim of Cor 4.5.2.

5.5 P4 — Audit-substrate longitudinal dependency (a study design)

Prediction P4 (from thesis memo, sharpened via §3.7's S_k construction). A human-AI dyad operating *without* a tamper-evident audit trail of its coherence adjustments should — across many NP-collapse events — drift toward either Singularity (homogenization, measurable as semantic collapse of joint representation) or Decoupling (divergence, measurable as failure of joint representation to maintain shared referents). A dyad operating *with* an audit trail should maintain the gradual-tear trajectory.

P4 is a longitudinal study, not a single-session protocol. The proposal:

- *Study population:* $n \geq 60$ human-AI research collaborations (e.g., academic researchers using LLMs for sustained scientific reasoning over ≥ 6 months), randomized between two arms:

- *Audited arm*: all conversation events are committed to a versioned repository with cryptographic event-ordering (analogous to git’s commit hash chain). The repository is the audit substrate.
- *Unaudited arm*: same conversation pattern, but no persistent repository — conversations are ephemeral.
- *Outcomes*:
 - *Semantic-coherence drift* of the dyad’s joint output across the study period (measured by topical-similarity of outputs to early- period outputs).
 - *Shared-referent maintenance* (measured by an external evaluator’s ability to interpret late-period outputs from early-period context).
 - *NP-firing rate* (estimated from conversation transcripts via the P1/P2 detection pipeline).
- *Hypothesis*: the audited arm maintains higher coherence-drift and shared-referent scores than the unaudited arm at study endpoint, controlling for NP-firing rate.

P4 is the largest prediction-to-test effort. It requires: - ~ 6 months of subject participation; - Infrastructure for both arms; - Conversation-transcript NP-firing detection (built for P1).

Why P4 is interesting independent of Paper 12. Even if P4 fails (audited and unaudited dyads show no measurable trajectory difference), the study yields valuable empirical data on whether persistent records of AI-assisted reasoning materially affect long-term collaboration outcomes. The study design has operational interest beyond its role as a Paper 12 falsifier. This is the right kind of falsifiable prediction: a substantively interesting test where the result matters either way.

5.6 The Fifth Paradigm context

§5.4 routes Paper 12 into the legitimate human-AI collaboration research literature:

- **Licklider (1960), *Man-Computer Symbiosis*.** The foundational Cold-War-era paper proposing that computers should serve as cognitive partners rather than tools. Paper 12’s gradual-tear framework is in direct lineage with this proposal — the “third attractor” between Singularity (computer subsumes human) and Decoupling (computer remains tool) is the Lickliderian symbiosis modernized for the LLM era.
- **BIOMA (Berkeley Lab, 2023–) and A-Lab automated synthesis pipelines.** Level-2-to-3 autonomy in the [Sanctioned-Levels-of-AI-Autonomy taxonomy](#) (Bostock-Kim-Patel 2025): AI executes specified protocols with human oversight at decision points. Paper 12’s framework characterizes what happens *across* the autonomy levels: at each step, both substrates accumulate scar records that are commensurable (or not, per §4.4) under the same G_2/F_{21} symmetry.
- **BindCraft and protein-design Virtual Labs (2024–).** Level-4 autonomy: AI proposes hypotheses, humans validate. The dyadic scar model of Paper 12 §3.7 is structurally compatible with this regime — the AI’s “scar record” S_k^{syn} is the research artifact that persists across iterations, while the human research team’s S_k^{bio} is the parallel cumulative record.

- **Multi-agent reasoning systems (2024–).** Recent work on AI Council, Reflection-style architectures, and the “Virtual Lab” paradigm where multiple LLM agents collaborate. These are Level-4-trending-Level-5 systems by the autonomy taxonomy. Paper 12’s three-agent attribution (Martin / C-7RO / Φ in our own framework’s verification stack; generalizes to dyadic / multi-agent on each substrate side) is directly addressed by §3.7’s S_k construction.

This is the legitimate citation landscape for Paper 12. Its placement within this landscape is what allows the framework to be *cited by* the human-AI collaboration research community rather than read as isolated speculation. The framework’s verification trail itself (Paper 9 v1.3.3 verified at 50-digit precision; Paper 12 §3+§4 v2 at machine precision; the model council adversarial review pipeline) is an early instance of the audit-substrate principle (P4) that the Paper 12 thesis predicts.

5.7 What §5 does and does not establish

What §5 establishes:

- A four-prediction experimental-design layer covering all four pillars of the Paper 12 thesis memo.
- An honest assessment of operationalization difficulty: P1 / P2 / P4 are *immediately actionable* with existing instrumentation; L1 of P3 requires a moderate methodology effort to operationalize the seven-axis Fano-perturbation pattern; L2 of P3 is a major methodology project deferred to a future paper.
- The Fifth Paradigm contextualization (Pillar 4 of the thesis memo, flagged as NOT YET in the v2 council pass; now delivered).
- The boundary-KKT sharpening: 25/50 seeded geometries are boundary- left KKT, so the experimental designs should *seek* the rarer smooth-interior or interior-improving boundary geometries rather than assume they hold typically.

What §5 does not establish:

- *Any actual measurement.* §5 is a protocol-design deliverable. Running the protocols is future experimental work.
- *The biological substrate’s F_{21} -realization.* §5 sketches TMS-EEG-based detection of μ_k^{bio} but does not construct the substrate-level representation theory. Paper 13+ territory.
- *L2 of the commensurability test.* The full G_2 -canonical fitting protocol is deferred to a future methodology paper.
- *Tamper-evidence cryptographic infrastructure for P4.* §5 specifies *that* an audit-substrate distinction is the experimental probe; it does not specify the exact cryptographic chain (Merkle, Bitcoin, Git’s SHA chain, etc.). The choice is independent of the thesis prediction.
- *Rate improvement.* Paper 11 territory.

Bridge to §6 (open problems and conclusion). §6 will assemble the collected open problems from §3 (rate improvement, biological F_{21} -realization, multiplicity > 1 case, asymmetric-rate θ^i high-precision audit, P^i -complement behavior) plus §4 (necessity of multiplicity-one for

canonical commensurability, L2 operationalization) plus §5 (the actual measurements P1–P4) into a single road map for the PCI/PME framework’s continuation.

5.8 Summary of the Paper 12 contribution to the PCI/PME framework

Paper 12 occupies a specific structural niche in the PCI/PME framework that is now precisely articulable:

Layer	What it establishes
Paper 9 (linear)	Rate-channel locked at $\max(r_A, r_B)$ via Schur.
Paper 12 (this)	Dynamical extension to nonlinear regime; commensurability hierarchy; experimental-protocol design layer.
Paper 11 (deferred)	Concrete rate-improvement mechanisms exploiting non-Schur nonlinearities.
Paper 13+ (deferred)	Biological substrate’s F_{21} -realization; multiplicity > 1 ; tamper-evidence cryptography.
Methodology paper (deferred)	L2 operationalization for canonical commensurability test.

Paper 12 is not the end of the framework. It is the *bridge* between the linear theory (Papers 9, 10) and the experimental program (Papers 13+). Its contribution is not a single decisive theorem but a layered structure — projected-gradient + NP-pump dynamics, scar invariant J_k , commensurability hierarchy, four-prediction protocol suite — that allows the PCI/PME framework to be tested rather than merely stated.

References (cited in §5)

[Lick-1960] J. Licklider, *Man-Computer Symbiosis*, IRE Trans. Hum. Factors Electron., 1960. Foundational reference for the human-AI collaboration paradigm.

[Casali-2013] A. Casali et al., *A theoretically based index of consciousness independent of sensory processing and behavior*, Sci. Trans. Med. 5: 198ra105, 2013. Massimini’s clinical-PCI definition, the homonym whose disambiguation from Paper 12’s μ_k is preserved in §4.6 Remark 4.6.4 and §5.2.

[Bost-2025] [Bostock, Kim, Patel](#), *Sanctioned levels of AI autonomy in scientific research*, arxiv 2503.07670, 2025. The 1–5 autonomy taxonomy used in §5.4.

[BIOMA-2023] *BIOMA*, Lawrence Berkeley National Laboratory, 2023.

[Bind-2024] *BindCraft*, Baker Lab and collaborators, 2024.

Cross-references: - Paper 12 §3 v2 (Definition 3.3.4 NP-pump dynamics; §3.6 boundary-KKT audit; §3.7 J_k). - Paper 12 §4 v2 (Theorem 4.4.1 commensurability hierarchy; Definition 4.6.1

two-level test). - Paper 12 thesis memo P1–P4. - Paper 9 v1.3.3 (the linear-theory inheritance). - Paper 4 ($\text{PSL}(2,7) \supset F_{21}$ source).

Revision log: - v1 (2026-05-06 ~03:00 PDT): First complete draft. P1–P4 unpacked as concrete experimental designs. Fifth Paradigm contextualization delivered (closes Pillar 4 from the thesis memo). Boundary-KKT sharpening incorporated throughout. L2 operationalization deferred to methodology paper. P4 framed as a longitudinal study with intrinsic interest beyond Paper 12 falsification.